STRATHMORE UNIVERSITY

SCHOOL OF COMPUTING AND ENGINEERING SCIENCES

MACHINE LEARNING-BASED TOOL FOR EARLY DETECTION OF AT-RISK STUDENTS IN HIGHER EDUCATION: CASE OF STRATHMORE UNIVERSITY

BY

KATHEMBO TSONGO DIEUDONNE

112721

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DECLARATION

Student's Declaration

I declare that this thesis is my original work and has not been presented to any other institution for examination or any other award. Where other sources of information have been used, they have been properly acknowledged.

Name: Kathembo Tsongo Dieudonne

Registration Number: 112721

Signature: _____ **Date:** _____

Supervisor's Declaration

This thesis has been submitted for examination with my approval as the University Supervisor.

Name: Dr. [Supervisor Name]

Title: [Lecturer/Senior Lecturer]

Department: [Department Name]

Signature: _____ **Date:** _____

Contents

Declaration **ii**

Table of Abbreviations 1

1 INTRODUCTION **2**

1.1 Background to the Study 2

1.2 Statement of the Problem 4

1.3 Research Objectives 6

1.3.1 Main Objective 6

1.3.2 Specific Objectives 6

1.4 Research Questions 6

1.5 Justification 7

1.5.1 Operational Necessity 7

1.5.2 Addressing the "Black Box" Problem 7

1.5.3 Policy and Economic Pressures 7

1.6 Scope and Delimitations of the Study 8

1.6.1 Scope 8

1.6.2 Delimitations 8

1.7 Definition of Terms 9

1.8 Significance 11

1.8.1 Theoretical and Methodological Significance 11

1.8.2 Practical Significance 12

1.8.3 National and Policy Significance 12

1.9 Summary 12

2 LITERATURE REVIEW **14**

2.1 Introduction 14

2.2 Global Perspectives on Student Attrition 14

2.3 Theoretical Frameworks for Student Retention 16

2.3.1 Tinto's Student Integration Model 16

2.3.1.1 The Longitudinal Mechanism 17

2.3.1.2 The Two Pillars of Integration 19

2.3.1.3 Limitations and Critiques 19

2.3.1.4 Operationalization in This Study 20

2.3.2 Astin's Input-Environment-Output (I-E-O) Model 20

2.3.2.1 Structure of the Model 20

2.3.2.2 Methodological Logic: The Natural Experiment 22

2.3.2.3 Complementarity with Tinto's Model 22

2.3.2.4 Operationalization in This Study 23

2.3.2.5 Limitations and Critiques of the I-E-O Model 24

2.4 Machine Learning in Educational Data Mining (EDM) 25

2.4.1 Overview of the Field 25

2.4.2 Core Applications in Higher Education 25

2.4.3 Machine Learning Algorithms and Techniques 26

2.4.4 Data Sources and Feature Engineering 27

2.4.5 Explainability, Fairness, and Ethical Considerations 27

2.4.6 Existing ML-based predictive analytics for at-risk students . . . 29

2.4.6.1 Georgia State University: GPS Advising System . . . 29

2.4.6.2 Morocco: A Modular and Explainable Machine Learning Pipeline at Mohammed V University 32

2.4.6.3 South Africa: Machine Learning for Engineering Student Success at the University of Johannesburg 36

2.4.6.4 South Africa: Bayesian Network Classification for Computer Science Student Risk Prediction at the University of Cape Town 39
2.4.6.5 Ethiopia: Predictive Analytics at Scale with African Institutional Data 42
2.4.6.6 Kenya: Business Intelligence and Decision Tree Classification for Student Retention Prediction —case of Machakos University College 43
2.4.6.7 Research Gaps Table 47
2.4.6.8 Relevance to our study 49
2.4.6.9 Conceptual Framework 51

Table of Abbreviations

Abbreviation	Description
AMS	Academic Management System
CBET	Competency-Based Education and Training
CESA	Continental Education Strategy for Africa (2016-2025)
EDM	Educational Data Mining
GPA	Grade Point Average
LMS	Learning Management System
ML	Machine Learning
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
SHAP	SHapley Additive exPlanations
SIM	Student Integration Model
SVM	Support Vector Machines

Chapter 1

INTRODUCTION

1.1 Background to the Study

Student retention and academic success represent critical challenges confronting higher education institutions worldwide. According to (Irwin et al., 2024), undergraduate enrollment in degree-granting institutions decreased by 13% between 2012 and 2022, while 23.3% of undergraduate students leave universities annually. Six-year graduation rates reveal substantial variation: 63% at public universities, 68% at private nonprofit institutions, and only 29% at private for-profit institutions. This persistent attrition represents significant personal and economic costs for individual students, alongside substantial challenges for institutional sustainability and mission fulfillment.

Technology is changing that. Machine learning algorithms can now identify at-risk students weeks or even months before traditional warning signs appear, achieving prediction accuracies of 88–92% (Ahmed, 2024; Gathani et al., 2026). Georgia State University's GPS Advising system illustrates what becomes possible: by tracking over 800 risk factors for 40,000 students daily and generating 90,000 targeted interventions each year, the university raised its graduation rate by 7.8 percentage points and virtually eliminated achievement gaps between demographic groups (Georgia State University, 2024). The technology, in short, is mature and proven—but its benefits have not yet reached most of the world.

Africa is a striking example of that gap. The African Union's Continental Education Strategy (CESA 2016–2025) acknowledges retention as a priority, yet most African universities still rely on end-of-semester transcripts as their primary diagnostic tool. Some

sub-regions have begun to experiment. Mohammed V University in Morocco achieved an 81.66% dropout-prediction accuracy in a pilot study that utilized L2-regularized logistic regression (Bouihi et al., 2024). "Similarly, a study at the University of Johannesburg in South Africa utilized machine learning to predict the performance of 1,366 engineering students, achieving a 65.86% accuracy rate using Decision Tree algorithms and identifying prior performance in physics and mathematics as primary success factors (M. Taodzera, 2016).

Within East Africa, Kenya's higher education sector faces particularly acute retention pressures. Government capitation covers only 57% of students against an 80% target, while public universities have accumulated Ksh 62 billion in unpaid bills—structural constraints that make student attrition both more likely and more financially consequential for students, families, and institutions (Daily Nation, 2024; Kenya Institute for Public Policy Research and Analysis, 2024).

Recognizing the challenge of student attrition, researchers from the University of Nairobi in collaboration with one researcher at the Free University of Brussels conducted a study utilizing Machakos University College as a case study to apply Business Intelligence and machine learning to dropout prediction (J. K. Ngemu et al., 2015). Using the WEKA toolkit to analyze historical institutional data, the research team—comprising Ngemu, Odongo, Omulo, and Manderick—evaluated multiple classification algorithms, including Naïve Bayes, Support Vector Machines (SVM), and the J48 decision tree. Their findings identified that students at-risk of dropping out could be accurately flagged based on critical predictors such as financial stability (fee payment), entry age, health status, and parental occupation (J. K. Ngemu et al., 2015).

Strathmore University operates within this challenging global landscape. While maintaining distinct academic standards that intensify the consequences of delayed intervention. The institution enforces a 67% minimum attendance threshold for examination eligibility and a 2.0 GPA floor for good academic standing—policy thresholds that render early identification especially critical, as students who breach them may be barred

from examinations or placed on academic probation regardless of their potential for recovery. The university collects extensive student data through its Academic Management System (AMS), Learning Management System (LMS), and attendance tracking platform, generating daily records on academic performance, engagement behaviour, and class participation.

Despite this data richness, these systems operate as isolated administrative repositories with no integrated analytics capability, preventing the transformation of raw institutional data into actionable early warnings. Students are typically identified for intervention only after manifesting metrics confirming trends and outcomes after events have occured. This reactive paradigm, though common across Kenyan institutions, is inconsistent both with international best practice (Georgia State University, 2024) and with Kenya's national call for greater technology utilisation in higher education (Republic of Kenya Ministry of Education, 2024). This study addresses that gap directly: by developing a machine learning-based predictive analytics platform calibrated to Strathmore's academic policies, data infrastructure, and resource constraints, it seeks to transform the institution's support model from reactive monitoring to proactive, evidence-driven intervention.

1.2 Statement of the Problem

Strathmore University currently faces a critical gap between data collection and actionable student support. Although the institution maintains robust data streams through its Academic Management System (AMS), Learning Management System (LMS), and attendance tracking platforms, these systems operate as isolated administrative repositories rather than integrated predictive tools. This fragmentation prevents the university from transforming raw institutional data into early warnings for at-risk undergraduate students.

The primary consequence of this fragmentation is a reactive support paradigm that carries measurable human costs. Students are typically identified for intervention only

after manifesting indicators of failure—cumulative low grades, breach of the mandatory 67% attendance threshold for examination eligibility, or formal withdrawal notice. By this stage, the window for effective academic recovery has often closed, and the outcomes that follow are both predictable and preventable: students drop out of programmes they could have completed, fail courses that timely tutoring referrals could have salvaged, and accumulate delays that extend their studies by entire semesters—each outcome representing a personal and institutional loss that earlier detection would have averted.

Despite possessing the necessary digital infrastructure and longitudinal student datasets, Strathmore University lacks a specialised machine learning architecture capable of integrating these data streams into a coherent risk stratification system. This absence is consequential: without the capability to detect deteriorating engagement trajectories weeks before they manifest as failed grades or breached attendance thresholds, the institution cannot intervene at the point when intervention is still effective. Without an automated ML-based tool to predict dropout risk, course failure risk, and program delay risk simultaneously, the university remains unable to allocate its limited counselling and mentoring resources to the students who need them most—and advisors are left to act on intuition rather than evidence, on crisis rather than prevention.

This study therefore develops, implements, and evaluates an integrated predictive analytics ML-based tool tailored to Strathmore University's academic policies, data infrastructure, and resource constraints. By producing explainable and actionable risk predictions across three distinct outcomes, the ML-based tool is designed to transition the institution from reactive to proactive monitoring and data-driven intervention.

1.3 Research Objectives

1.3.1 Main Objective

To design, develop, and evaluate a machine learning-based predictive analytics tool for early identification of at-risk undergraduate students at Strathmore University.

1.3.2 Specific Objectives

1. To design a Machine Learning predictive tool for university administrators, school administrators, and academic mentors that present actionable risk predictions with supporting intervention guidance.
2. To evaluate model accuracy and predictive performance using historical student cohort data (2021-2024) and standard machine learning evaluation metrics including Accuracy, Precision, Recall, F1-score, and ROC-AUC.
3. To assess the platform's usability and perceived institutional impact through structured evaluation with academic advisors, school administrators, and mentors, with specific attention to intervention feasibility and long-term sustainability.

1.4 Research Questions

1. How can risk predictions be effectively presented to different stakeholder groups—university administrators, school administrators, and academic mentors—to support informed decision-making?
2. To what extent does training predictive models on historical student cohort data (2021-2024) with known outcomes produce valid and generalisable predictions for current enrolled students?
3. How do academic advisors, school administrators, and mentors perceive the platform's usability, intervention feasibility, and potential for long-term institutional

adoption?

1.5 Justification

1.5.1 Operational Necessity

This research is justified by the need to bridge the gap between data collection and data utilization in Universities such as Strathmore University. Currently, Strathmore's student support is reactive—interventions. It often occurs only after a student has already failed or dropped out. There is an urgent need for a system that transforms underutilized data into actionable early warnings without increasing the administrative burden on faculty.

1.5.2 Addressing the "Black Box" Problem

This study is justified by its focus on explainability. By using specialized techniques to show why a student is at risk, the research justifies the shift toward evidence-based mentoring, ensuring that institutional support is credible.

1.5.3 Policy and Economic Pressures

This study responds to the Kenya Ministry of Education's Sessional Paper No. 1 of 2024, which cites inadequate technology use as a core bottleneck in higher education (Kenya Institute for Public Policy Research and Analysis, 2024). In a climate of rising enrollments and limited government funding, the status quo of universal mentoring is no longer sustainable. This research is justified as an economic necessity to ensure that finite counseling resources are directed only where they are most needed.

1.6 Scope and Delimitations of the Study

1.6.1 Scope

This research focuses on the development and evaluation of a predictive analytics platform for approximately 5,000 undergraduate students across all five schools at Strathmore University, Kenya. The technical scope involves integrating data from the Academic Management System (AMS), Learning Management System (LMS), and attendance tracking records into a unified analytics pipeline. The study develops three prediction models—targeting dropout risk, course failure risk, and program delay risk—using ensemble algorithms including Random Forest, Gradient Boosting, and XGBoost, trained on historical student cohort data from 2021-2024 with known academic outcomes. Model predictions are accompanied by SHAP-based explainability outputs to ensure advisors and administrators can understand and act on the reasons behind each risk classification (Lundberg & Lee, 2017). The resulting platform includes role-based dashboards for university administrators, school administrators, and academic mentors, with model performance measured through standard machine learning evaluation metrics (Accuracy, Precision, Recall, F1-score, and ROC-AUC) and platform effectiveness assessed through structured stakeholder feedback.

1.6.2 Delimitations

The research boundaries are defined by the following deliberate constraints. Regarding data factors, the analysis is restricted to academic risk factors measurable through institutional systems, such as attendance rates, GPA, LMS engagement, and course grades. External variables such as socioeconomic background, mental health status, and family circumstances are excluded, as they are not systematically captured in Strathmore's existing data infrastructure and their collection would raise ethical concerns beyond the current scope. In terms of student population, the study is limited to undergraduate students across Strathmore University's five schools; postgraduate and continuing

education cohorts are excluded due to their unique assessment patterns.

The system capabilities are delimited to identifying risk levels and suggesting general intervention categories, such as academic counselling or tutoring referrals, rather than providing course-specific remediation. Within the algorithmic scope, the study employs ensemble tree-based methods like Random Forest and XGBoost (Chen & Guestrin, 2016) rather than deep learning architectures. This decision is justified by the dataset size of approximately 5,000 students and the critical need for feature interpretability. Furthermore, the technical platform is developed as a web-based application, with mobile-native development remaining outside the scope. Financial domains, including fee payment risk and scholarship management, are also excluded to focus purely on academic and engagement indicators. Finally, the implementation phase concludes with a functional prototype and structured evaluation; full-scale production deployment and long-term longitudinal impact assessment are reserved for future research.

1.7 Definition of Terms

Term	**Description**
At-Risk Student	A student whose data patterns statistically indicate a high probability of dropout, course failure, or program delay prior to the actual event.
Dropout Risk	The predicted probability of a student discontinuing enrolment before completing their degree requirements.
Course Failure Risk	The predicted likelihood of a student failing a course or becoming exam-ineligible due to attendance falling below the mandatory 67% threshold at Strathmore University.
Program Delay Risk	The probability that a student will exceed the standard programme duration due to academic probation, course repetition, or accumulated failed semesters.

Term	Description
Academic Management System	The primary institutional database for official student records, including demographic information, enrolment status, and academic history.
Learning Management System	A digital platform used to deliver course content and track student engagement through behavioural data such as login frequency, resource access, and assignment submissions.
Early Warning System	An integrated tool that uses data and predictive modelling to trigger timely alerts for students identified as requiring academic intervention.
Engagement Metrics	Quantitative measures of student participation derived from institutional systems, such as LMS login frequency, assignment submission timeliness, and attendance rate trajectories.
Ensemble Learning	A machine learning technique that combines multiple algorithms—such as Random Forest, Gradient Boosting, and XGBoost—to produce predictions that are more accurate and stable than any single model alone.
Feature Engineering	The process of transforming raw institutional data into optimised input variables for machine learning model training, including the creation of derived indicators such as attendance flags and GPA thresholds.
Intervention	A proactive support action—such as academic mentoring, tutoring referral, or advisor outreach—triggered by a student's identified risk status and designed to prevent academic failure or withdrawal.
Predictive Analytics	A branch of data analytics that uses statistical modelling and machine learning techniques applied to historical data to forecast future outcomes.

Term	Description
Explainability	The capacity of a predictive model to provide human-interpretable reasons for its predictions, often implemented through SHAP values.
Historical Cohort	A defined group of students whose complete academic trajectories—from enrolment to graduation—are known and documented.
Model Accuracy	The proportion of correct predictions (both at-risk and not-at-risk) made by a model relative to total observations.
Precision	The proportion of students predicted as at-risk who are genuinely at-risk. High precision minimises false alarms.
Recall (Sensitivity)	The proportion of genuinely at-risk students who are correctly identified by the model.
F1-Score	The harmonic mean of Precision and Recall, providing a single balanced metric.
ROC-AUC	A threshold-independent metric measuring a model's ability to discriminate between classes. A value of 1.0 indicates perfect discrimination.

1.8 Significance

1.8.1 Theoretical and Methodological Significance

This study advances the field of Educational Data Mining (EDM) by operationalizing Tinto's Student Integration Model within an African institutional context. It provides empirical evidence that Western retention theories can be mapped to quantifiable data elements. Methodologically, it demonstrates that SHAP-based explainability is not a luxury but a prerequisite for AI adoption in resource-constrained environments, setting a new standard for how predictive models should be evaluated beyond simple accuracy.

1.8.2 Practical Significance

For Institutions: The study provides a documented, open-source blueprint for transitioning from reactive to proactive support. It offers a sustainable alternative to the individual-dependency models that have failed in other regional universities by embedding automated data pipelines into the university's core infrastructure.

For Educators: The platform empowers advisors with role-based dashboards that move decision-making from intuition to evidence. It transforms the advisor's role from a monitor to a facilitator, allowing for substantive, reason-led conversations based on specific risk drivers.

For Students: As the primary beneficiaries, students gain a safety net. The system transforms punitive policies (like the 67% attendance rule) into supportive frameworks, allowing for academic recovery before thresholds are irreversibly breached.

1.8.3 National and Policy Significance

At the national level, this research serves as a replicable model for the Republic of Kenya's transition to Competency-Based Education and Training (CBET). By demonstrating how a mid-sized university can implement high-level analytics within tight budget constraints, the study provides a policy instrument for other East African institutions to improve retention rates without the prohibitive costs of commercial software.

1.9 Summary

This chapter established the context, rationale, and direction of the study. Beginning at the global level, it demonstrated that student dropout remains a critical challenge in higher education and that machine learning-based predictive analytics has proven capable of identifying at-risk students weeks before conventional indicators surface—yet these benefits remain largely inaccessible to number of African universities.

The logic of disparate systems forces institutions like Strathmore into a reactive sup-

port paradigm where students are identified only after breaching the 67% attendance threshold or 2.0 GPA floor. This gap, situated within Kenya's national call for greater technology utilization in education, defines the problem this study addresses.

The study responds by developing, ML-based predictive analytics platform tailored to Strathmore's policies and resource context. The platform predicts dropout, course failure, and programme delay using ensemble algorithms trained on the 2021–2024 historical cohort. Its contributions span theory, practice, and policy: advancing retention frameworks in African contexts, providing a sustainable and replicable institutional tool, and demonstrating a cost-effective model for comparable universities across the region. Chapter Two now reviews the literature underpinning these contributions, examining theoretical frameworks, documented system implementations with their successes and limitations, and the precise knowledge gaps this study is designed to fill.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

The purpose of this chapter is to synthesize the multidisciplinary body of research that underpins this study's design and motivation. The review spans three interconnected domains: the theoretical frameworks governing student retention, the global evolution of predictive analytics and Educational Data Mining (EDM) in higher education, and the African and Kenyan institutional contexts in which this research is embedded. Together, these streams establish both the intellectual foundation and the practical necessity of a machine learning-based Early Warning System (EWS) calibrated to Strathmore University's academic policies and data infrastructure.

The chapter is organized to trace a deliberate logical arc. First, it begins with the macroeconomic and human costs of student attrition. Second, it narrows progressively from global best practices to regional pilot studies. And finally, it focuses on the specific institutional gaps this research is designed to fill. Central to this arc is the argument that modern machine learning does not merely add technological convenience; it operationalizes classical retention theory, transforming abstract constructs such as academic integration and social engagement into measurable and actionable data signals.

2.2 Global Perspectives on Student Attrition

Student retention is a global imperative linked directly to both institutional sustainability and the fulfillment of higher education's social mission. According to Irwin et al. (2024), undergraduate enrollment in degree-granting institutions in the United States

decreased by 13% between 2012 and 2022, compounded by a persistent annual attrition rate of 23.3%. Six-year graduation rates reveal substantial structural disparities: 68% at private nonprofit institutions, 63% at public universities, and only 29% at private for-profit institutions. Despite a modest recent recovery—the 2024 National Student Clearinghouse report recorded a national first-year retention rate of 68%, described as the highest in the past decade (National Student Clearinghouse Research Center, 2024)—the structural problem of attrition remains unresolved, with large proportions of students continuing to exit before degree completion.

These figures carry dual consequences. At the institutional level, each departing student represents lost tuition revenue, wasted administrative investment, and diminished institutional capacity to fulfill its educational mission. At the individual level, students who withdraw often exit the system bearing financial debt without the credential necessary for socioeconomic mobility. A systematic review of 125 peer-reviewed studies on undergraduate retention published between 2014 and 2022 concluded that, despite decades of research and institutional intervention, student dropout remains a persistent and structurally significant problem across higher education globally, with research in the field remaining fragmented and without a unified evidence base for effective remediation (Gonçalves et al., 2024). The following table summarizes graduation rate variation by institution type:

Institution Type	**Six-Year Graduation Rate**
Private Nonprofit Institutions	68%
Public Universities	63%
Private For-Profit Institutions	29%

Table 2.1: Six-Year Graduation Rates by Institution Type (Irwin et al., 2024)

Beyond the United States, the problem is not geographically bounded. Widening access policies across the Global South have expanded enrollment without proportional improvements in completion, meaning that participation gains mask a deepening com-

pletion deficit (Gonçalves et al., 2024). In sub-Saharan Africa and Kenya specifically, this deficit is compounded by resource constraints, infrastructure gaps, and the financial pressures documented in Section 2.7, making the attrition problem both more acute and less resourced than in high-income institutional contexts. These global statistics therefore establish the urgency of the problem and demonstrate that student attrition is not a peripheral concern but a structural threat to the viability of higher education institutions worldwide—providing the macro-level context within which Kenya's institutional challenges, and this study's intervention, must be understood.

2.3 Theoretical Frameworks for Student Retention

2.3.1 Tinto's Student Integration Model

The dominant theoretical anchor for student retention research remains Vincent Tinto's Student Integration Model (SIM). First proposed by Tinto and Cullen in 1973 and formally published as a peer-reviewed synthesis in 1975, the model draws its conceptual foundations from Émile Durkheim's theory of suicide— specifically the proposition that a deficit of social cohesion precipitates withdrawal from the community (Tinto, 1975). Tinto's scholarly influence has been exceptional: his works have been cited in more than 700 studies across multiple disciplines, establishing the SIM as the most widely adopted theoretical lens in higher education retention research (Bento Júnior et al., 2025).

At its core, the SIM conceptualizes the university as a microcosm comprising two interdependent systems—the academic and the social—and posits that a student's persistence is governed by their degree of integration into both (Tinto, 1975, 1993). Critically, Tinto frames dropout not as a single discrete event but as a *longitudinal process*: the student's commitment to persist is continuously reconfigured through ongoing interactions. This interactional character means that dropout cannot be attributed to any one variable in isolation; it emerges from the cumulative weight of how individual at-

tributes, academic performance, and social relationships interact and accumulate over time (Nicoletti, 2019; Tinto, 1993).

2.3.1.1 The Longitudinal Mechanism

The SIM operates through a structured, multi-stage sequence linking pre-institutional characteristics to eventual persistence:

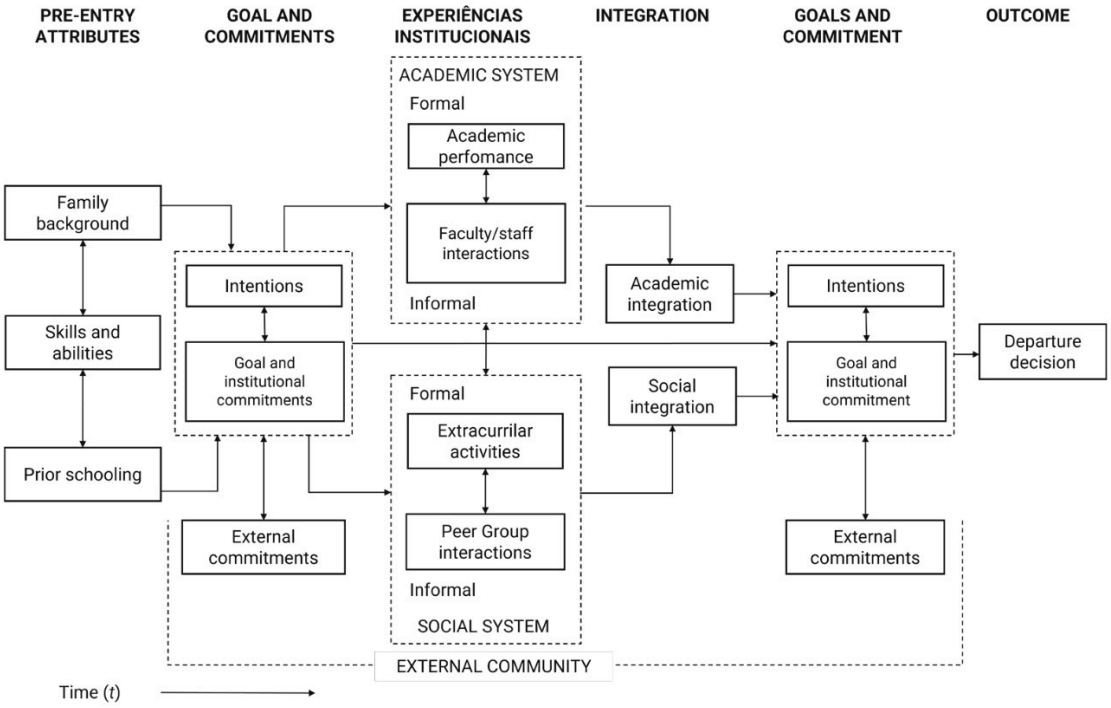

Figure 2.1: Tinto's Model of Institutional Departure

1. **Pre-Entry Attributes:** Students enter the institution carrying three foundational background variables: family background, individual skills and abilities, and prior schooling. These attributes do not directly determine persistence but collectively shape the intellectual and social disposition the student brings to the institution (Tinto, 1975).

2. **Initial Goals and Commitments:** Pre-entry attributes feed into two parallel commitment structures. *Intentions and goal commitments* reflect the student's desire to earn a degree, while *goal and institutional commitments* reflect the desire to graduate from a specific institution. Alongside these sits a third force—*external commitments*—representing obligations outside the institution, such as

family or employment, which compete with academic integration from the outset (Tinto, 1975, 1993).

3. **Institutional Experiences:** Once enrolled, students interact simultaneously with two systems. The *Academic System* operates through both formal channels—academic performance—and informal channels—faculty and staff interactions. The *Social System* operates through formal extracurricular activities and informal peer group interactions. Both systems are embedded within and influenced by the broader *External Community* (Tinto, 1993).

4. **Integration:** The quality of institutional experiences determines the degree of integration achieved along two dimensions. *Academic integration* results from engagement with the Academic System, while *social integration* results from engagement with the Social System. These two dimensions are interdependent—weakness in either increases the probability of departure (Nicoletti, 2019; Tinto, 1975).

5. **Re-evaluated Goals and Commitments:** Integration feeds back into a revised set of commitments—updated intentions, goal and institutional commitments, and external commitments—which are now reconfigured by the student's actual lived experience within the institution. This re-evaluation stage is what makes the model longitudinal and iterative rather than static: commitments are not fixed at entry but continuously reshaped over time (Nicoletti, 2019; Tinto, 1993).

6. **Departure Decision (Outcome):** The re-evaluated commitments produce the final outcome: a *departure decision*. If the cumulative weight of academic and social integration has reinforced the student's goal and institutional commitment, persistence follows. If integration has been insufficient and external commitments or personal costs outweigh the perceived benefits of continued enrollment, departure follows. Critically, the entire process unfolds across time (t), underscoring that dropout is a process, not a moment (Tinto, 1975).

2.3.1.2 The Two Pillars of Integration

Tinto distinguishes two primary and interdependent dimensions of integration, both of which must be achieved to sustain persistence (Tinto, 1993):

Academic Integration encompasses both structural and normative dimensions. Structural integration is measured by grade performance (GPA) relative to institutional standards, while normative integration reflects intellectual development and alignment with the academic community's values. Grades function not merely as performance records but as institutional signals of belonging within the academic system; a declining GPA is therefore not simply a performance metric but a symptom of deteriorating academic integration (Bento Júnior et al., 2025; Tinto, 1975).

Social Integration reflects the quality of peer-group interactions and faculty relationships. Research on blended and online learning environments has confirmed that when instructors provided timely support, appropriate feedback, and interactive engagement, students were substantially more likely to persist (Ivankova & Stick, 2007). Furthermore, studies measuring LMS engagement have established that completers consistently demonstrate significantly higher levels of participation in learning activities—measured through discussion posts, resource access, and login frequency—than withdrawers, validating these behavioral signals as digital proxies for integration (Morris et al., 2005).

2.3.1.3 Limitations and Critiques

Despite its canonical status, the SIM carries limitations directly relevant to this study's context. Its key constructs—integration, commitment, fit—lack precise operational definitions, creating interpretive challenges when translating the model into computable variables (Nicoletti, 2019). The model was also developed within a Western, residential university context and requires careful contextual recalibration before application in non-Western institutions such as Strathmore University. Additionally, traditional SIM-based research has relied predominantly on retrospective, correlational designs: the model identifies *what* predicts departure but does not generate prospective, real-time

predictions for individual students. This retrospective limitation is the precise gap that machine learning addresses—where the SIM identifies the risk factors, ML determines when those factors are currently activating for a specific student.

2.3.1.4 Operationalization in This Study

This study operationalizes Tinto's model within Strathmore's digital infrastructure by systematically mapping each theoretical construct onto a measurable institutional variable. The mapping is deliberate and bidirectional: every predictor variable selected for the ML models corresponds to a specific SIM construct, ensuring that algorithmic outputs retain theoretical interpretability.

2.3.2 Astin's Input-Environment-Output (I-E-O) Model

Alongside Tinto's SIM, Alexander Astin's Input-Environment-Output (I-E-O) model provides a complementary and practically powerful theoretical lens for this study. Originally developed as a conceptual guide for institutional assessment, the model was empirically consolidated in Astin's landmark study *What Matters in College? Four Critical Years Revisited*, which drew on surveys of more than 20,000 students, 25,000 faculty members, and 200 institutions to examine how academic programmes, faculty, peer groups, and institutional policies shape student development and outcomes (Astin, 1993). The model's philosophical foundation is a *talent development* approach to institutional excellence: quality is not measured by institutional prestige or resource endowment, but by the degree to which the institution succeeds in developing the potential of the students it admits (Astin & Antonio, 2012). This orientation makes the I-E-O model especially relevant to the Strathmore context, where the goal is not to rank students but to identify and support those whose potential is at risk of being lost.

2.3.2.1 Structure of the Model

The I-E-O model organizes the variables relevant to student success into three sequential, interacting components (Astin & Antonio, 2012):

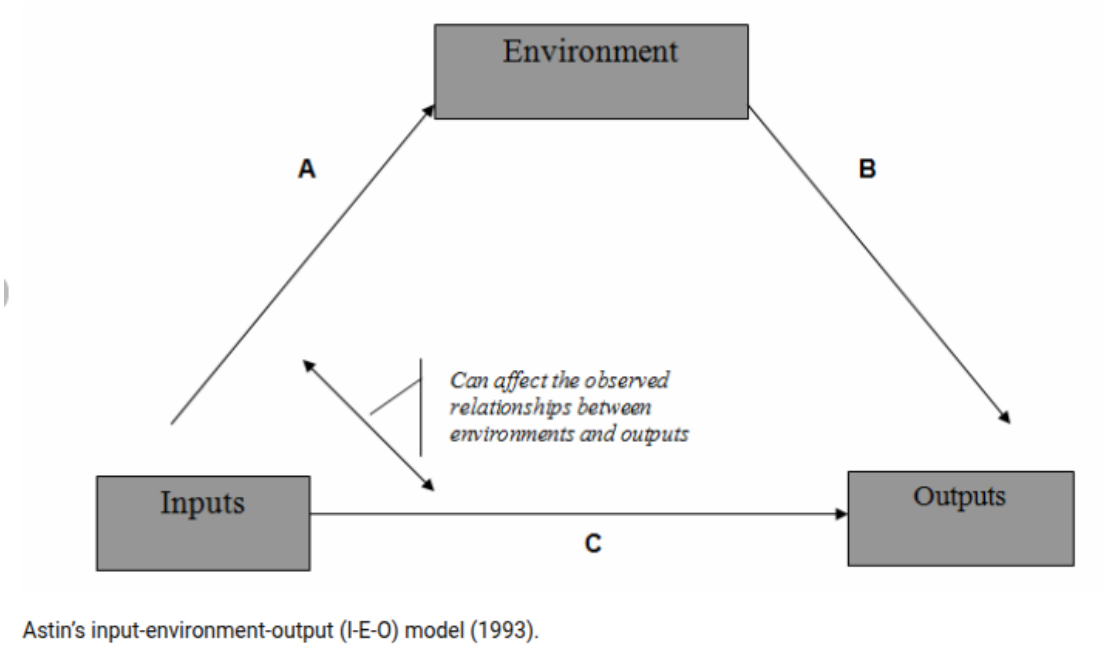

Figure 2.2: Astin's input-environment-output (I-E-O) model (1993).

1. **Inputs** refer to the personal qualities and characteristics students bring with them at the point of entry. This includes demographic attributes (age, sex), family background, prior academic achievement such as secondary school GPA, and initial motivations and goals. Astin's framework treats no single input as deterministic; rather, inputs establish the baseline against which the effects of institutional experiences must be measured (Astin & Antonio, 2012).

2. **Environment** encompasses the full range of institutional experiences, programmes, policies, and interactions to which the student is exposed during enrollment. This includes both tangible elements (classrooms, library resources, residential halls) and intangible elements (teaching pedagogy, administrative policies, peer group composition, and the quality of faculty interaction). Environmental variables are particularly significant in the I-E-O framework because they represent the factors that educators and institutional administrators can directly control to maximize talent development (Astin & Antonio, 2012). Applied research using this framework has confirmed that student-advisor interaction, student-faculty interaction, extracurricular participation, and library utilization are each positively and inde-

pendently associated with first-to-second year retention (Xiao et al., 2020).

3. **Outputs** (or Outcomes) represent the measurable results of the educational programme: academic achievement, retention, degree completion, and the full range of skills, values, and attitudes the institution aims to develop. In the I-E-O framework, outcomes are understood as the product of how environmental conditions transform a student's initial input characteristics. Evaluating outcomes without accounting for inputs is explicitly identified by Astin as an incomplete assessment design, as it fails to capture how much students actually developed relative to their starting point (Astin & Antonio, 2012).

2.3.2.2 Methodological Logic: The Natural Experiment

A defining methodological feature of the I-E-O model is its use of what Astin terms *natural experiments*. Because students cannot be randomly assigned to different institutional environments, the model relies on complex multivariate statistical analyses—such as multiple logistic regression—to control for input differences and isolate the true effect of environmental variables on outcomes (Astin & Antonio, 2012). This approach offers two practical advantages. First, it avoids the artificial conditions of laboratory experiments by studying real institutional variation. Second, it allows researchers to examine the simultaneous effects of many environmental variables, which is precisely the analytical challenge confronted in this study's multi-stream data pipeline. The machine learning models developed in Chapter Three operationalize this same logic: by training on historical student cohort data with known outcomes (2021–2024), the models statistically isolate which combinations of input and environmental variables most strongly predict each risk outcome, mirroring Astin's natural experiment methodology in a computational framework.

2.3.2.3 Complementarity with Tinto's Model

The I-E-O model and Tinto's SIM are theoretically complementary rather than competing. Tinto explains *why* students depart through the mechanism of diminishing integra-

tion; Astin provides the *variable architecture*— the structured categorization of inputs, environmental conditions, and outcomes—that a predictive model must measure (Bento Júnior et al., 2025; Nicoletti, 2019). This complementarity is particularly significant for the three-outcome prediction framework of this study. Dropout risk maps most directly to Tinto's integration failure mechanism. Course failure risk and programme delay risk, however, are better understood through Astin's environmental lens: they reflect whether the institution's academic policies, attendance requirements, and support resources are functioning as effective environmental buffers against disengagement. A student may remain enrolled— not a dropout in Tinto's terms—while still accumulating course failures or programme delays that Astin's output variables would capture.

2.3.2.4 Operationalization in This Study

The I-E-O model maps directly and systematically onto the three institutional data streams integrated in this study's analytics pipeline:

Inputs are operationalized through Academic Management System (AMS) variables: entry qualifications, prior GPA trajectory, school of enrollment, and year of study. These correspond to Astin's input variables and establish the baseline characteristics against which environmental effects are assessed (Astin & Antonio, 2012).

Environmental variables are operationalized through LMS engagement metrics—login frequency, resource access, and assignment submission timeliness—and real-time attendance records. These capture the student's degree of active interaction with the institutional environment. Empirical research applying the I-E-O framework has demonstrated that student-advisor interaction, student-faculty interaction, extracurricular participation, and library utilization are each positively associated with first-to-second year retention (Xiao et al., 2020), directly validating the use of engagement metrics as environmental proxies in this study. Strathmore's specific institutional policies—the 67% attendance threshold and the 2.0 GPA floor— function as measurable environmental constraints that the predictive models are explicitly calibrated to reflect.

Outputs are the three predicted risk outcomes: dropout risk, course failure risk, and programme delay risk. By framing these as distinct output categories, this study extends Astin's output framework beyond the traditional graduation/dropout binary to capture the graduated spectrum of academic difficulty that students actually experience. This extension is methodologically consistent with Astin's own recognition that institutional outcomes are multidimensional and that talent development can be undermined in forms that fall short of full withdrawal (Astin & Antonio, 2012).

2.3.2.5 Limitations and Critiques of the I-E-O Model

Despite its widespread adoption, the I-E-O model carries documented limitations that must be acknowledged. The model was developed within a Western, residential, four-year university context, where full-time on-campus participation is assumed to be the primary channel of environmental engagement (Astin, 1993). This assumption does not transfer without recalibration to commuter, part-time, or non-Western institutional settings (Karp et al., 2008).

A further limitation is the model's linear input-to-environment-to-output structure, which oversimplifies the recursive nature of student development. In practice, outputs such as academic performance continuously reshape subsequent environmental engagement, creating feedback loops that a static framework cannot capture (Astin & Antonio, 2012). These limitations do not invalidate the I-E-O model as a theoretical anchor for this study; rather, they motivate the adoption of a longitudinal machine learning architecture that continuously re-evaluates risk as new engagement and performance data arrives, compensating for the model's structural simplifications in a real institutional deployment context.

2.4 Machine Learning in Educational Data Mining (EDM)

2.4.1 Overview of the Field

Educational Data Mining (EDM) is a rapidly developing interdisciplinary field that investigates techniques for identifying significant patterns in large-scale educational datasets to enhance teaching, learning, and institutional efficacy (Bello et al., 2025). Within this domain, Machine Learning (ML) has become a foundational tool, shifting EDM from theoretical investigation into a practical instrument for data-driven decision-making. By applying computational analysis to data originating in educational contexts—such as student information systems, Learning Management Systems (LMS), and behavioral interaction logs—ML algorithms identify hidden patterns to forecast future student performance and behavior (Dirin & Saballe, 2022). A recent systematic review of 72 peer-reviewed studies concluded that research in this domain has matured significantly, with ensemble methods, deep learning, and explainable AI now representing the three dominant technical paradigms (Aldowah et al., 2026).

2.4.2 Core Applications in Higher Education

The primary applications of machine learning in EDM converge on four objectives that are directly relevant to this study:

Performance Prediction involves forecasting academic outcomes— such as final course grades, semester GPA, or examination results—by analyzing historical academic records, engagement trajectories, and behavioral patterns. These models allow institutions to move beyond end-of-term assessment toward continuous, data-informed monitoring of student progress (Kwakye, 2025).

Risk Identification applies ML models to flag at-risk students early in a term, enabling timely and targeted interventions before failure becomes irreversible. This is the primary objective of the Early Warning System (EWS) architecture developed in this study (Bello et al., 2025).

Study Path and Programme Guidance uses algorithmic predictions of student preferences, strengths, and motivational orientations to recommend appropriate academic specializations. Research at Haaga-Helia University demonstrated that Random Forest (94% accuracy) and Decision Tree (93% accuracy) are effective classifiers for study path selection, outperforming logistic regression when non-linear motivational and demographic features are included (Dirin & Saballe, 2022).

Adaptive and Personalized Learning supports the creation of intelligent learning environments that tailor content, pacing, and feedback to individual student profiles based on continuous engagement monitoring (Kwakye, 2025).

2.4.3 Machine Learning Algorithms and Techniques

Supervised learning is the most widely adopted paradigm in EDM, primarily applied to classification tasks—predicting categorical outcomes such as pass/fail, dropout/persist, or risk level—and regression tasks that produce continuous scores (Bello et al., 2025; Dirin & Saballe, 2022).

Ensemble Methods such as Random Forest, XGBoost, and Gradient Boosting are consistently identified as top performers in EDM contexts due to their ability to manage complex, non-linear relationships in educational data more effectively than baseline linear models. Bello et al. (2025) demonstrated this directly: on a dataset of 1,163 student records, a baseline Linear Regression model achieved an R^2 of only .32, while Random Forest achieved .69 and XGBoost achieved .70, confirming the superior generalization capacity of ensemble methods (Bello et al., 2025).

Deep Learning models—including Multi-Layer Perceptrons (MLP), Convolutional Neural Networks (CNN), and Long Short-Term Memory (LSTM) networks—are increasingly utilized to capture intricate temporal interactions between features. While these architectures often achieve high predictive accuracy, they operate as "black boxes," providing limited insight into the reasoning behind individual predictions, which significantly limits their institutional adoption in advisory contexts (Aldowah et al., 2026;

Kwakye, 2025).

Traditional Classifiers including Decision Trees, Support Vector Machines (SVM), Naïve Bayes, and Logistic Regression remain widely used for their relative simplicity, computational efficiency, and inherent interpretability. Their performance, however, is generally outpaced by ensemble methods when the underlying data contains non-linear feature interactions (Bello et al., 2025).

2.4.4 Data Sources and Feature Engineering

ML models in EDM rely on diverse institutional data sources, including Academic Management Systems, Learning Management Systems, and clickstream logs from digital platforms (Kwakye, 2025). Feature engineering is a critical intermediate step, transforming raw institutional records into structured model inputs such as attendance rates, prior grade trajectories, assignment submission timeliness, and temporal engagement patterns.

Research has established that *time-based features*—such as the intervals between LMS interactions, the recency of last login, and the "burstiness" of activity relative to assignment deadlines—carry significant predictive power, often capturing early disengagement signals that static features such as a single GPA value cannot detect (Dirin & Saballe, 2022). To ensure model stability and address the class imbalance that characterizes most student risk datasets—where at-risk students constitute a minority—standard pre-processing techniques including normalization, one-hot encoding, and the Synthetic Minority Oversampling Technique (SMOTE) are routinely applied (Aldowah et al., 2026; Bello et al., 2025).

2.4.5 Explainability, Fairness, and Ethical Considerations

As ML becomes embedded in institutional decision-making, three interrelated concerns—fairness, accountability, and transparency— have emerged as critical prerequisites for responsible deployment (Bello et al., 2025; Kwakye, 2025).

The Black Box Problem and Explainable AI (XAI): Complex ensemble and deep learning models produce predictions without human-interpretable reasoning, creating an institutional trust deficit that prevents adoption by advisors and administrators who cannot interrogate or defend the model's outputs (Kwakye, 2025). To address this, researchers now employ XAI techniques—most prominently SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), and Partial Dependence Plots (PDP)—to provide feature-level explanations for individual predictions (Kwakye, 2025). A Springer study applying CatBoost–SHAP to student academic crisis warning demonstrated that explainable predictions provide "intuitive decision support and guidance assistance for education administrators," validating SHAP's role as a trust-enabling layer in institutional deployments (Zhang & Tao, 2021).

Bias and Fairness: Predictive models trained on historically biased institutional data risk reinforcing systemic inequalities, particularly for students from disadvantaged socioeconomic backgrounds. Without deliberate fairness auditing, a model may achieve high aggregate accuracy while systematically misclassifying vulnerable subpopulations (Bello et al., 2025).

Human-in-the-Loop Deployment: Ethical best practice consistently emphasizes that ML models should function as decision-*support* tools that complement rather than replace human pedagogical judgment (Kwakye, 2025). The advisor remains the responsible agent; the model supplies evidence. This principle directly informs the role-based dashboard architecture of this study, where SHAP outputs are presented as advisor-facing explanations rather than autonomous risk verdicts.

Data Privacy: Institutional deployment of student analytics systems requires compliance with data protection frameworks—including GDPR in European contexts and equivalent national regulations— mandating robust data anonymization, access controls, and transparent data governance policies (Kwakye, 2025).

2.4.6 Existing ML-based predictive analytics for at-risk students

2.4.6.1 Georgia State University: GPS Advising System

Institutional Context and Motivation

Georgia State University (GSU) is a large public research university in Atlanta (Georgia State University, 2024). It enrolls more than 50,000 students—of whom 58% qualify as low-income by federal standards, 60% are non-white, and 30% are first-generation college students—provides the most extensively documented and empirically evaluated implementation of ML-based predictive analytics for student retention in higher education (Georgia State University, 2024; Renick, 2017). Prior to 2012, GSU's annual dropout rate stood at approximately 5,600 students per year, with adviser-to-student ratios as high as 1:1,000. Students—particularly those from minority and low-income backgrounds—were failing and withdrawing without ever seeking help, because the institution had no mechanism for identifying them before failure had already occurred (Kamenetz, 2016).

System Architecture and Methodology

In 2011, GSU entered a partnership with the Education Advisory Board (EAB)—a specialist higher education analytics consultancy—to develop what would become the GPS (Graduation and Progression Success) Advising system (Kamenetz, 2016; Wan, 2019). The methodology proceeded in three phases:

Phase 1 — Historical Data Analysis: EAB analysts processed ten years of GSU institutional records, encompassing 2.5 million course grades and 140,000 student records, to identify patterns statistically associated with student attrition and programme delay (Association of Governing Boards, 2019; Kamenetz, 2016).

Phase 2 — Risk Factor Development: From this historical analysis, a library of more than 800 risk markers was developed. These markers span a wide range of behavioral and academic signals: course registration anomalies (such as a senior enrolling in a course that does not count toward their degree), grade patterns, financial account status,

LMS engagement changes, and attendance indicators derived from Wi-Fi and LMS login monitoring (EdTech Magazine, 2019; Georgia State University, 2024). Risk markers range from *repeated historical patterns*—behaviours that have historically preceded dropout across thousands of prior students—to *specific points of change*, such as a sudden drop in grades or a significant reduction in LMS usage (Georgia State University, 2024).

Phase 3 — Operational Deployment: The GPS system went live in August 2012. It operates as a nightly batch process: student grades, records, and behavioral indicators are updated every night, and the system generates push-notification alerts to the assigned academic adviser whenever a student triggers one or more risk markers (Association of Governing Boards, 2019; Kamenetz, 2016). Advisers are expected to initiate contact with the flagged student within 48 hours (EdTech Magazine, 2019). The system is delivered through EAB's proprietary web platform, *Navigate*, which presents each student's risk score on a color-coded dashboard (green/yellow/red) alongside the specific alert triggers (The Signal, 2019).

Outcomes and Impact

The results of GPS Advising are among the most cited in the student success literature. Since its launch, GSU has recorded the following verified outcomes (Association of Governing Boards, 2019; Georgia State University, 2024; National Institute for Student Success, 2024):

- Graduation rates increased by 23 percentage points over the decade following implementation, with four-year graduation rates specifically improving by 7.8 percentage points.
- The system generates more than 90,000 adviser-initiated interventions per academic year, compared to approximately 1,000 annual adviser meetings before 2012.
- Over 250,000 one-on-one meetings between advisers and students have been

prompted by GPS alerts since 2012.

- Achievement gaps based on race, ethnicity, and income level have been eliminated: Black, Hispanic, first-generation, and low-income students now graduate at rates at or above the overall student body rate—the only national university in the United States to have achieved this outcome.
- A Boston Consulting Group analysis confirmed that the programme generates more revenue than it costs, with each one-percentage-point increase in retention generating approximately $3.18 million in additional tuition revenue annually (Wan, 2019).

Limitations and Critical Assessment

Despite its documented success, the GPS Advising system carries several significant limitations that are directly relevant to our study's design choices:

Proprietary and Non-Transferable Architecture: The most consequential limitation is that GPS Advising is built on EAB's proprietary *Navigate* platform, for which GSU paid a heavily discounted rate of $150,000 per year as an early adopter (Wan, 2019). Market-rate licensing costs are substantially higher for most institutions in the Global South, including Kenyan universities operating under severe resource constraints. This creates a fundamental equity problem: the institutions most in need of student success technology are precisely those least able to afford commercial vendor solutions.

Lack of Model Transparency: Because the underlying algorithmic logic of the *Navigate* platform is proprietary, institutions cannot inspect, audit, or modify the predictive models. This opacity prevents the kind of institution-specific calibration that this study argues is essential: Strathmore University's 67% attendance threshold and 2.0 GPA floor require that predictive models be explicitly tuned to local policy parameters, which is impossible within a black-box commercial system (The Signal, 2019; Wan, 2019).

Bias Risk from Historical Data: The risk markers were derived from historical insti-

tutional data that may encode pre-existing systemic inequalities. Scholars have raised concerns that predictive systems trained on historically biased data risk reinforcing those patterns, particularly when racial and socioeconomic variables are correlated with risk markers (Wan, 2019). Although GSU explicitly excludes race from its algorithmic inputs, proxy variables such as financial aid status and registration patterns may still function as demographic proxies (Wan, 2019).

Resource-Intensive Implementation: GSU's success was not solely the product of technology. More than 90% of the total programme cost was in staffing: GSU hired 42 new academic advisers in the first six months, reducing the adviser-to-student ratio from 1:1,000 to approximately 1:300 (Kamenetz, 2016). This level of staffing investment is not replicable in resource-constrained contexts, and it raises the question of whether the technology or the additional human capacity was the primary driver of improved outcomes.

Relevance to This Study

The Georgia State case establishes two foundational propositions for this research. First, it provides proof of concept at institutional scale: predictive analytics, when properly implemented and supported by human advisory capacity, can dramatically improve retention and eliminate equity gaps (Georgia State University, 2024). Second, and equally importantly, its limitations define the design requirements for the current study. The Strathmore tool intended to be developed in this thesis is therefore not simply inspired by the Georgia State model—it is explicitly designed to resolve the transferability and transparency gaps that prevent that model from being deployed in a Kenyan institutional context.

2.4.6.2 Morocco: A Modular and Explainable Machine Learning Pipeline at Mohammed V University

Institutional Context and Motivation

While the Georgia State GPS Advising system demonstrates the institutional impact

achievable through predictive analytics at scale, its proprietary architecture and resource requirements limit its transferability to institutions in the Global South. A critically important and more recent contribution to the EDM literature—directly relevant to the African higher education context—is a modular dropout prediction pipeline developed at the Mohammadia School of Engineers, Mohammed V University in Rabat, Morocco, by Bettahi, Belouadha, and Harroud (2025). This study is geographically proximate to the Kenyan context, uses open-source tools, and explicitly addresses the governance and explainability requirements that the GSU system does not (Bettahi et al., 2025).

Dataset and Institutional Setting

The study used a longitudinal institutional dataset of 17,883 undergraduate records drawn from multiple cohorts at a Moroccan HEI. After cleaning and feature curation, the modeling matrix comprised 91 features integrating four categories of student data: demographic and programme descriptors; academic progression variables (cumulative GPA, credits earned, course failures, and passed course counts); engagement and attendance metrics; and financial and administrative signals encoded as binary hold flags (`HAS_*` variables) with paired timestamp durations. The positive class—students recorded by the registrar as having discontinued their studies—comprised 2,856 students (16.0%), producing a substantially imbalanced classification problem typical of real institutional data (Bettahi et al., 2025).

Methodology and Pipeline Architecture

The pipeline was implemented entirely in **Python 3.12.0** using open-source libraries, specifically `scikit-learn 1.7.1`, `xgboost 3.0.3`, and `shap 0.48.0`. This is a significant methodological distinction from the proprietary EAB *Navigate* platform used by Georgia State: every component of the Moroccan pipeline is reproducible, auditable, and transferable without licensing costs (Bettahi et al., 2025).

The six-stage modular architecture proceeded as follows (Bettahi et al., 2025):

1. **Data Ingestion and Cleaning:** Automatic delimiter detection; removal of columns

with $> 90\%$ missingness; median imputation for numeric fields; label encoding for categorical variables with an explicit <MISSING> token; date normalization to elapsed days; and leakage prevention by dropping direct student identifiers.

2. **Feature Engineering:** Transformation of raw institutional records into 91 structured predictors, including temporal features such as hold durations (e.g., AP_END_DTE − AP_START_DTE) and derived academic momentum indicators.

3. **Train/Test Split:** A stratified 80/20 split yielding 14,306 training and 3,577 test students, preserving the empirical class ratio in both partitions.

4. **Model Benchmarking:** Nine algorithms were evaluated: Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), k-Nearest Neighbors (k-NN), Support Vector Machine (SVM), Gradient Boosting (GB), XGBoost, Naïve Bayes (NB), and Multilayer Perceptron (MLP).

5. **Explainability Layer (SHAP):** SHAP TreeExplainer applied to all tree-based models on the full 3,577-test-case set; KernelExplainer with k-means background ($k = 100$) applied to MLP on a 300-sample subset.

6. **Reporting and Visualization:** Structured output directories storing confusion matrices, ROC curves, SHAP plots, performance summaries, and reproducibility logs.

Model validation used stratified 5-fold cross-validation with a fixed random seed (42). A single operating threshold τ was selected per model by maximizing F1 on cross-validation folds and then fixed for all test and subgroup reporting (Bettahi et al., 2025).

Results and Performance

Table 2.2 summarizes the headline test-set performance of all nine models.

Model	Accuracy	Precision	Recall	F1	AUC-ROC
XGBoost	0.970	0.879	**0.944**	**0.911**	**0.993**
Random Forest	0.968	0.919	0.877	0.898	0.989
MLP	0.964	0.891	0.884	0.888	0.979
Gradient Boosting	0.963	0.909	0.855	0.881	0.988
Decision Tree	0.961	0.862	0.900	0.881	0.936
SVM	0.954	0.816	0.921	0.865	0.983
k-NN	0.948	0.892	0.767	0.825	0.947
Logistic Regression	0.917	0.682	0.904	0.777	0.968
Naïve Bayes	0.689	0.322	0.858	0.468	0.881

Table 2.2: Model performance on the Moroccan HEI test set (N=3,577; dropout rate 16%). Adapted from Bettahi et al. (2025).

XGBoost achieved the highest performance across all primary metrics: AUC-ROC of 0.993, F1-score of 0.911, and recall of 0.944. McNemar's paired significance testing confirmed that XGBoost was statistically superior to Gradient Boosting ($p = 0.008$) on the test set, while the difference from Random Forest was not statistically significant ($p = 0.428$) (Bettahi et al., 2025). To assess generalizability, the full pipeline was replicated on the public UCI Portugal dataset (Realinho et al., 2021), where XGBoost remained top-ranked (F1-score = 0.792, AUC-ROC = 0.922), confirming model ordering preservation across feature-shifted datasets.

Limitations

The authors explicitly acknowledge several limitations directly relevant to this study's design (Bettahi et al., 2025):

Institution-Specific Features: Administrative hold timestamps such as `AP_START_DTE` and `AP_END_DTE` are Moroccan HEI-specific and may not exist or carry equivalent semantics in other institutional contexts. This motivates this study's institution-specific feature engineering strategy, which maps Strathmore's 67% attendance threshold and

2.0 GPA floor as policy-calibrated features rather than relying on transferable generic variables.

Batch Processing: The pipeline is designed as a static train/test evaluation rather than a real-time streaming system. Deployment as a live Early Warning System would require architectural extension to support continuous model retraining as new semester data arrives.

Relevance to This Study

The Bettahi et al. (2025) pipeline is the most methodologically aligned existing system in the literature relative to the platform developed in this thesis. Three direct connections are drawn. First, the use of Python with open-source libraries (`scikit-learn`, `xgboost`, `shap`) establishes the technical feasibility of deploying a high-performance, explainable EWS without commercial licensing—directly addressing the transferability limitation of the Georgia State model. Second, the confirmation of XGBoost as the dominant algorithm for structured, imbalanced HEI data on an African institutional dataset validates the algorithm selection rationale for this study. Third, the SHAP-based risk versus protective factor distinction provides the explainability architecture template for the role-based advisor dashboards developed in Chapter Four. The key departure of this study is then the simultaneous prediction of three distinct risk outcomes—dropout risk, course failure risk, and programme delay risk—rather than a single binary dropout outcome, and the explicit calibration of all models to Strathmore University's specific policy thresholds.

2.4.6.3 South Africa: Machine Learning for Engineering Student Success at the University of Johannesburg

One of the earliest investigations of ML-based student performance prediction in the South African context was conducted by Taodzera, Twala, and Carroll at the University of Johannesburg's Department of Electrical and Electronics Engineering Science (T. Taodzera et al., 2016). South Africa presents a uniquely complex setting for ed-

ucational data mining: engineering programmes face critically low throughput rates, students arrive with profoundly unequal secondary school preparation resulting from the enduring effects of apartheid-era education, and socioeconomic disparity produces wide variation in academic readiness within the same classroom (T. Taodzera et al., 2016).

Dataset and Features

The study worked with 1,366 engineering student records drawn from a South African university. Available features spanned demographic and academic entry indicators: high school Mathematics score, Physics score, English score, Life Orientation score, ethnicity, home language, age, school province, and religion. The response variable was defined as the number of subjects passed to qualify for progression to second year—a binary pass/fail classification motivated by the institution's own progression regulations (T. Taodzera et al., 2016).

Feature Selection

Correlation analysis identified Physics score (r = 0.2391) and Mathematics score (r = 0.1836) as the strongest predictors of first-year success, followed by age (r = 0.1191) and ethnicity (r = 0.0808). The significance of ethnicity as a predictor directly reflects structural inequalities in South African secondary schooling, where historically advantaged schools continue to produce students substantially better prepared for the quantitative demands of engineering study (T. Taodzera et al., 2016). Wrapper-based feature selection confirmed Mathematics and Physics as the primary predictors, with ethnicity and school province also identified as significant—the latter functioning as an indirect proxy for geographic inequality and school system quality differentials across provinces.

Models and Results

Five algorithms were evaluated: Logistic Regression (LR), Naïve Bayes (NB), Artificial Neural Network (ANN), Support Vector Machine (SVM), and Decision Tree (DT). The

DT algorithm performed best with 65.86% accuracy, an F1-score of 0.658, and a Kappa statistic of 0.316—classifying it as a fair predictor. SVM ranked second (64.63%), followed by NB (63.41%) and LR (63.17%). ANN performed weakest (60.24%). All models were significantly better than the prior South African study by Mashiloane (2015), which failed to exceed 50% accuracy on a Computer Science dataset from the University of the Witwatersrand (T. Taodzera et al., 2016).

Critical Assessment

The Taodzera et al. study makes two contributions of direct relevance to this thesis. First, it demonstrates that standard ML algorithms can be operationalized in a resource-limited African institutional setting using only enrollment-time features available at the point of admission—without requiring LMS integration, real-time data pipelines, or proprietary vendor platforms. Second, the identification of school province and ethnicity as statistically significant predictors highlights a structural feature of African educational data that does not appear in the North American literature: that secondary schooling inequality is itself a measurable risk factor that encodes into university-level performance data.

However, the study carries significant limitations. The overall accuracy ceiling of 65.86% reflects the constraints of using enrollment-time data only, without any longitudinal academic performance features. The absence of SHAP-based or any form of explainability analysis means the model cannot be operationalized as a policy-actionable early warning system: advisers cannot understand which specific factors triggered a risk flag for an individual student. Furthermore, the modest Kappa values (all below 0.32) suggest that all five models classify at a level only slightly above what chance would produce, and none would meet the precision threshold required for institutional deployment without substantial risk of false positives that misdirect support resources (T. Taodzera et al., 2016).

2.4.6.4 South Africa: Bayesian Network Classification for Computer Science Student Risk Prediction at the University of Cape Town

At the University of Cape Town's Department of Computer Science, Hassen conducted a study applying Bayesian Network classifiers to predict which second-year Computer Science students that were at risk of failing CSC2001F (Hassen, 2020). This was a core data structures course widely regarded as the critical attrition point in the UCT Computer Science major. The institutional motivation was explicit: while first-year pass rates remained comparatively stable, the transition to senior courses caused a substantial proportion of students to either exceed the standard three-year degree duration or abandon the Computer Science major entirely (Hassen, 2020). This pattern of second-year attrition, compounded by South Africa's post-apartheid imperative to produce technically skilled graduates from historically disadvantaged backgrounds, provided the normative urgency for the study.

Dataset and Features

The study drew on 728 student records spanning six cohorts from 2013 to 2019. Features were selected through iterative consultation with the course convenor and spanned both pre-entry and early in-course indicators: Matric Mathematics mark, Matric English mark, National Benchmark Test (NBT) scores in Mathematics, Quantitative Literacy and Academic Literacy, first-year Computer Science course final marks (CSC1015F and CSC1016S), the first assignment mark for CSC2001F, and the CSC2001F Test 1 mark (Hassen, 2020). The target variable was binarised as Pass (final mark above 49) or Fail (final mark below 50). Several candidate features — including UCTScore and first-year GPA — were ultimately excluded, the former because its inclusion would have reduced the usable dataset by approximately ten percent due to missingness (Hassen, 2020).

Methodological Design

Three Bayesian Network models were developed using the Netica platform. The first

was manually constructed from expert knowledge gathered in faculty consultations. The second was generated automatically using Netica's Tree Augmented Naïve Bayes (TAN) structure learning algorithm, which approximates variable interactions through a tree structure imposed on a naïve Bayesian base (Hassen, 2020). The third was a combined model synthesising insights from both, with direct links to the AtRisk target node drawn from whichever predecessor model demonstrated greater sensitivity for each predictor. The gradient descent algorithm was selected for conditional probability table learning in both the TAN and combined models following comparative evaluation of three available learning procedures.

A notable methodological contribution was the adoption of incremental training, wherein each cohort's model was trained exclusively on all preceding years' data and evaluated on the subsequent cohort — replicating the temporal constraints of genuine deployment (Hassen, 2020). This approach avoids the data leakage that afflicts many EDM studies that train and test on cross-sectional splits of the same cohort. Additionally, class imbalance — a persistent challenge in student risk datasets where passing students substantially outnumber failing students — was addressed through SMOTE oversampling, and continuous features were discretised into uniform ten-point intervals in accordance with EDM literature recommending discretisation for improved Bayesian classifier accuracy (Hassen, 2020).

Models and Results

The three models produced substantially different performance profiles when evaluated on the 2019 holdout cohort (n=108). The manually constructed model performed poorly, achieving only a 67.59% accuracy rate alongside a recall of 0.33 and a near-negligible F1 score of 0.10, indicating that expert intuition alone was insufficient to capture the probabilistic dependencies embedded in the data (Hassen, 2020). The TAN model improved substantially on accuracy (84.25%) and recall (0.50), though its precision remained low at 0.17. The combined model produced the most instructive outcome: while its accuracy (83.33%) was marginally below the TAN model's, it

achieved a perfect recall score of 1.0 — correctly identifying every at-risk student in the test cohort — alongside an F1 score of 0.40 and a precision of 0.25 (Hassen, 2020). The study's most important finding was the identification of CSC2001F Assignment 1, CSC2001F Test 1, CSC1016S final mark and Matric Mathematics as the variables exerting the greatest influence on the AtRisk node, as determined through Netica's variance reduction sensitivity analysis.

Critical Assessment

The Hassen study makes several contributions of direct relevance to this thesis. First, it demonstrates that Bayesian Network classifiers can be operationalised effectively within a South African Computer Science institutional context using a modest dataset without requiring large-scale LMS integration or commercial predictive analytics platforms. Second, the adoption of incremental cohort-based training represents a methodologically sound simulation of real-world deployment that is absent from many comparable EDM studies. Third, the explicit prioritisation of recall over accuracy as the primary evaluation criterion is theoretically well-motivated in the student support context: the asymmetric cost of missing a genuinely at-risk student substantially outweighs the cost of providing unnecessary support to a student who would have passed regardless (Hassen, 2020).

However, the study carries significant limitations that circumscribe its generalisability. The dataset of 728 records spread across six cohorts is small by machine learning standards, and the per-cohort test sets are consequently limited — the 2019 evaluation cohort contained only 108 students, of whom just six were classified as failing. This severe class imbalance, even after SMOTE augmentation of the training sets, means that the combined model's perfect recall of 1.0 must be interpreted cautiously: correctly identifying all six failing students in a 108-student cohort is an encouraging result but constitutes insufficient evidence of robust generalisability across larger or differently composed populations (Hassen, 2020). The precision score of 0.25 further reveals that for every genuine at-risk student correctly flagged, three false positives were generated

— an operational burden that could misallocate scarce advising resources and undermine institutional confidence in the system.

The study also notably excludes social and demographic features on ethical grounds, explicitly acknowledging that school-level quality and apartheid-era structural inequality are likely significant unobserved confounders (Hassen, 2020). While this decision is ethically defensible, it means the model cannot surface the structural drivers of differential risk that are most actionable at the policy level. Furthermore, the absence of any explainability framework beyond Netica's sensitivity analysis means that, while a course convenor can observe that Assignment 1 is the strongest predictor, they cannot generate student-level explanations sufficient to guide individualised advising conversations.

2.4.6.5 Ethiopia: Predictive Analytics at Scale with African Institutional Data

A more recent and methodologically stronger study from the East African context was conducted by Esmael (2024) at Wollo University and the Kombolcha Institute of Technology in Ethiopia (Esmael, 2024). Using a dataset of 32,582 students from academic years 2017–2022 extracted from the institutional student information system, this study applied ML algorithms including Random Forest, Gradient Boosting, and Decision Tree classifiers to predict final academic results from demographic and academic entry features: gender, region, entrance examination results, number of previous attempts, studied credits, and disability status. Python was used for all preprocessing and modeling.

The study found that ensemble methods—particularly gradient boosting—outperformed single classifiers on Ethiopian institutional data, echoing the pattern observed in both the Moroccan Bettahi et al. (2025) pipeline and the global EDM literature. Entrance examination results and studied credits emerged as the strongest predictors, consistent with findings from both the South African and Moroccan contexts that prior academic achievement and engagement intensity are the most transferable cross-institutional risk signals (Esmael, 2024).

This study's principal contribution to the present literature review is confirmation that ensemble tree-based methods retain their predictive advantage over classical algorithms even on African institutional data with limited feature dimensionality. Its limitation is the absence of any explainability layer (no SHAP or LIME analysis is conducted), meaning that model outputs cannot be directly translated into student-specific intervention recommendations (Esmael, 2024).

2.4.6.6 Kenya: Business Intelligence and Decision Tree Classification for Student Retention Prediction —case of Machakos University College

One of the few published East African studies applying machine learning techniques to student retention prediction in a higher education context was conducted by Ngemu, Odongo, Omulo, and Manderick, drawing on data from Machakos University College in Kenya (J. M. Ngemu et al., 2015). The study was situated within a broader business intelligence (BI) framing, positioning educational data mining not merely as an academic exercise but as a strategic institutional tool for evidence-based decision making in student management. The motivating concern was the well-documented but institutionally underaddressed problem of student withdrawal in Kenyan higher learning institutions, where personal, financial, and socioeconomic factors converge to produce attrition patterns that existing student information systems lacked the analytical capacity to anticipate or explain (J. M. Ngemu et al., 2015).

Dataset and Features

The study drew on secondary survey data sourced from Machakos University College's institutional database, comprising 270 instances and 14 attributes. The feature set was notable for its emphasis on socioeconomic and demographic indicators rather than prior academic performance: variables included course enrolled, KCSE grade, gender, father's and mother's educational qualifications, difficulty in fees payment, parental occupation, marital status of parents, sponsor or guardian type, age on entry, whether course expectations were met, health status, and course-programme match (J. M. Ngemu et al., 2015). The binary target variable distinguished between students classified as PERSIST

or DROPOUT. The deliberate inclusion of family financial circumstances, parental employment status, and student health as primary predictors reflects the Kenyan institutional context, where economic precarity and household-level resource constraints constitute the dominant structural drivers of non-completion rather than the academic preparation deficits more characteristic of South African engineering attrition (J. M. Ngemu et al., 2015).

Feature Selection

Attribute importance analysis was conducted using both information gain and gain ratio measures within the WEKA environment, with Ranker search used to order predictors by significance. Both evaluators converged on the same top four predictors: difficulty in fees payment (DFP), age on entry (AOE), parental occupation (PO), and student health status (HTH) (J. M. Ngemu et al., 2015). The dominance of fees payment difficulty as the single most informative predictor — with a gain ratio of 0.424, substantially exceeding the second-ranked predictor at 0.153 — is a finding of considerable structural significance. It positions financial exclusion, rather than academic underpreparation, as the primary mechanism of student attrition in this Kenyan institutional setting, a pattern consistent with broader literature on non-completion in sub-Saharan African higher education where fee-dependent public universities serve students from predominantly low-income households (J. M. Ngemu et al., 2015). Gender and course type ranked lowest among predictors in both evaluators, suggesting that the withdrawal risk in this population was predominantly socioeconomic rather than programme-specific or gender-differentiated.

Models and Results

Four classification algorithms were evaluated using WEKA: the J48 decision tree (an implementation of C4.5), Naïve Bayes, Multilayer Perceptron (ANN), and Support Vector Machine (SMO). All four were assessed on ten-fold cross-validation accuracy and recall, as well as model building execution time (J. M. Ngemu et al., 2015). The J48 decision tree outperformed all competitors, achieving a cross-validation accuracy of

94.8% and a recall of 94.4%, while also being the fastest to train at 0.02 seconds. Multilayer Perceptron ranked second in accuracy (93.4%) but required more than two seconds to train — over 100 times slower than J48. Naïve Bayes and SVM both achieved approximately 90% accuracy and recall. On the training set, J48 correctly classified 153 of 162 instances with an accuracy of 94.4% and a Kappa statistic of 0.872. On the holdout test set, J48 classified 78 of 81 instances correctly, yielding a test accuracy of 96.3% and a Kappa of 0.925 (J. M. Ngemu et al., 2015). The decision tree structure surfaced interpretable classification rules, including the finding that fee payment difficulty alone was sufficient to classify a student as DROPOUT, while a student with no fee difficulties and good health was classified as PERSIST regardless of other attributes (J. M. Ngemu et al., 2015).

Critical Assessment

The Ngemu et al. study makes several contributions relevant to this thesis and to the broader East African educational data mining literature. First, it represents one of the earliest published attempts to apply supervised classification methods to student retention prediction within a Kenyan public university, demonstrating that WEKA-based BI tools can be operationalised using institutional administrative data without requiring specialist data infrastructure or proprietary analytics platforms (J. M. Ngemu et al., 2015). Second, the feature set's emphasis on socioeconomic and household-level variables rather than academic performance metrics is theoretically well-grounded in Tinto's model of student departure and its African adaptations, and produces a predictor profile that is meaningfully distinct from both the South African engineering context of Taodzera et al. and the Computer Science course performance context of Hassen. Third, the decision tree output generates directly interpretable classification rules that an institutional registrar or student affairs officer could apply without specialist data science literacy, addressing a practical barrier to deployment that more opaque models such as ANNs and SVMs do not overcome.

However, the study carries critical methodological limitations that substantially circum-

scribe the interpretability and generalisability of its reported results. The most significant concern is the extreme smallness of the dataset: 270 instances with 14 attributes is a very limited sample for constructing robust machine learning models, and the ten-fold cross-validation accuracy of 94.8% and test set accuracy of 96.3% are almost certainly inflated by overfitting to a dataset too small to capture genuine population-level variation in attrition patterns (J. M. Ngemu et al., 2015). The Kappa statistic of 0.872 on the training set and 0.925 on the test set suggest strong agreement, but these figures must be treated with caution given the dataset size. The study does not report precision and recall separately for each class — a critical omission when the class distribution between PERSIST and DROPOUT students is unknown and potentially imbalanced, since high overall accuracy can mask poor identification of the minority dropout class in exactly the manner the authors of subsequent studies such as Hassen explicitly sought to avoid (J. M. Ngemu et al., 2015).

Furthermore, the study provides no discussion of class imbalance, no oversampling or resampling adjustment for minority class representation, and no SHAP-based or equivalent explainability analysis beyond the tree structure itself. The absence of any longitudinal or cohort-based evaluation means the reported accuracy figures reflect model performance on a single static cross-section of institutional data, rather than the temporally realistic scenario in which a model trained on prior cohorts is used to flag at-risk students in a new intake. The feature set, while theoretically motivated, relies entirely on enrollment-time socioeconomic and demographic data and includes no in-course academic performance indicators — meaning the model cannot be updated or refined as early course performance data becomes available, limiting its utility as a dynamic early warning system. Finally, the authors acknowledge that social and cultural factors could not be collected, and do not discuss the ethical implications of using parental occupation, family financial status, and health as institutional risk flags, dimensions of the deployment context that are increasingly recognised as requiring explicit equity and bias auditing before any model is operationalised for advising or support allocation decisions (J. M. Ngemu et al., 2015).

2.4.6.7 Research Gaps Table

Study	Context	Technique Used	Key Predictors	Identified Gaps
Georgia State University GPS Advising (Georgia State University, 2024)	USA, 50,000+ students	800+ risk markers; proprietary EAB Navigate platform	Grades, LMS engagement, financial status, registration anomalies	Proprietary and non-transferable; no model transparency; resource-intensive staffing; unaffordable for Global South institutions
Bettahi et al. (Bettahi et al., 2025)	Morocco, 17,883 undergraduate records	XGBoost, Random Forest, SHAP explainability	GPA, credits earned, course failures, financial hold flags	Static batch pipeline; no real-time retraining; institution-specific features limit transferability; single binary dropout outcome only
Taodzera et al. (T. Taodzera et al., 2016)	South Africa, 1,366 engineering students	Decision Tree, SVM, Naïve Bayes, ANN, Logistic Regression	Physics score, Mathematics score, age, ethnicity	Enrollment-time data only; no longitudinal features; no explainability layer; Kappa values below 0.32 indicate near-chance classification

Study	Context	Technique Used	Key Predictors	Identified Gaps
Hassen (Hassen, 2020)	South Africa, 728 CS students (UCT)	Bayesian Network (manual, TAN, combined)	CSC2001F Assignment 1, Test 1, CSC1016S mark, Matric Mathematics	Very small dataset; high false positive rate (precision = 0.25); no SHAP-based explainability; limited generalisability beyond single course
Esmael (Esmael, 2024)	Ethiopia, 32,582 students	Random Forest, Gradient Boosting, Decision Tree	Entrance exam results, studied credits, gender, region	No explainability layer; no SHAP or LIME analysis; cannot generate student-specific intervention recommendations
Ngemu et al. (J. M. Ngemu et al., 2015)	Kenya, 270 instances (Machakos University College)	J48 Decision Tree, Naïve Bayes, SVM, MLP (WEKA)	Fee payment difficulty, age on entry, parental occupation, health status	Extremely small dataset; accuracy likely inflated by overfitting; no class imbalance handling; no per-class precision/recall; enrollment-time features only; no explainability

Table 2.3: Summary of Research Gaps in Existing ML-Based Student Attrition Studies

2.4.6.8 Relevance to our study

Study	Contribution to This Study	Design Element Adopted	Design Element Improved or Departed From
Georgia State University GPS Advising (Georgia State University, 2024)	Proof of concept that predictive analytics at scale can raise graduation rates and eliminate equity gaps	Role-based dashboards; frequenty risk score updates; adviser-initiated intervention model	Replaced proprietary platform with open-source pipeline; calibrated to Strathmore-specific thresholds (67% attendance, 2.0 GPA); eliminated vendor dependency
Bettahi et al. (Bettahi et al., 2025)	Most methodologically aligned study; validates XGBoost and SHAP for African HEI dropout prediction using open-source tools	Python open-source stack (scikit-learn, XGBoost, SHAP); stratified train/test split; SHAP risk vs. protective factor visualisation	Extended from single binary dropout outcome to three simultaneous outcomes (dropout, course failure, programme delay); added Strathmore-specific policy features

Study	Contribution to This Study	Design Element Adopted	Design Element Improved or Departed From
Taodzera et al. (T. Taodzera et al., 2016)	Demonstrates ML operationalisation in a resource-limited African setting; highlights structural inequality as a measurable risk factor	Confirms feasibility of African institutional ML deployment without proprietary infrastructure	Added longitudinal LMS and attendance features beyond enrollment-time data; incorporated SHAP explainability; targeted higher accuracy ceiling
Hassen (Hassen, 2020)	Validates incremental cohort-based training as deployment-realistic evaluation; establishes recall as the primary metric in student support contexts	Incremental training logic; prioritisation of recall over accuracy; SMOTE oversampling for class imbalance	Replaced Bayesian Networks with ensemble methods (Random Forest, XGBoost); added SHAP explainability for student-level advisor guidance
Esmael (Esmael, 2024)	Confirms ensemble methods outperform classical classifiers on African institutional data even with limited feature dimensionality	Validates Random Forest and Gradient Boosting as primary algorithm candidates for African HEI contexts	Added SHAP explainability layer; extended feature set to include LMS engagement and attendance beyond demographic and entry variables

Study	Contribution to This Study	Design Element Adopted	Design Element Improved or Departed From
Ngemu et al. (J. M. Ngemu et al., 2015)	Earliest Kenyan institutional ML study; establishes financial exclusion as the dominant attrition driver in Kenyan public universities	Theoretical grounding of socioeconomic risk factors in Tinto's model; confirms feasibility of Kenyan institutional deployment	Replaced WEKA with Python ensemble pipeline; addressed class imbalance via SMOTE; added SHAP explainability; incorporated in-course academic and engagement features; used substantially larger dataset

Table 2.4: Relevance of Reviewed Studies to our study's Design

2.4.6.9 Conceptual Framework

Framework Explanation

The solution architecture presented in Figure **??** describes a nine-stage end-to-end pipeline through which raw institutional data is transformed into actionable, explainable risk predictions for Strathmore University's academic support staff. The framework flows strictly from top to bottom, with each stage building on the output of the preceding one.

Stage 1 — Data Sources: The pipeline originates from three institutional systems operating in parallel: the Academic Management System (AMS), which supplies enrollment records, GPA trajectories, and academic standing data; the Learning Management System (LMS), which contributes behavioral engagement signals including login frequency, resource access, and assignment submission patterns; and the Attendance Tracking Platform, which provides real-time attendance rates and flags students

approaching the mandatory 67% examination eligibility threshold.

Stage 2 — ETL Pipeline: Raw data extracted from the three source systems is passed through an Extract-Transform-Load (ETL) interface that merges records across systems, resolves inconsistencies, and validates data integrity before any modelling takes place.

Stage 3 — Feature Engineering: The cleaned dataset is transformed into a structured feature matrix. This stage applies normalization, categorical encoding, SMOTE oversampling to address class imbalance, and the construction of derived indicators such as attendance breach flags, GPA floor proximity scores, and temporal engagement patterns.

Stage 4 — Model Training: Three ensemble algorithms — Random Forest, XGBoost, and Gradient Boosting — are trained on the engineered feature matrix using historical student cohort data from 2021 to 2024. Stratified cross-validation and hyperparameter tuning are applied to maximise generalisation performance.

Stage 5 — Prediction Engine: The trained models generate three simultaneous risk scores for each currently enrolled student: dropout risk, course failure risk, and programme delay risk. This multi-outcome architecture reflects the graduated spectrum of academic difficulty that students experience, extending beyond the binary dropout classification common in prior literature.

Stage 6 — SHAP Explainability: Risk scores alone are insufficient for institutional deployment. This stage applies SHAP (SHapley Additive exPlanations) to decompose each prediction into its contributing feature drivers, identifying both risk factors and protective factors at the individual student level. This transforms opaque model outputs into human-interpretable explanations that advisors can act on.

Stage 7 — REST API: A role-gated REST API serves as the interface between the ML pipeline and the frontend dashboards, ensuring that each user role receives only the data and predictions appropriate to their institutional responsibilities.

Stage 8 — Role-Based Dashboards: Predictions and SHAP explanations are surfaced through three distinct dashboards calibrated to different institutional roles. The University Administrator dashboard provides a university-wide retention overview and escalation controls. The School Administrator dashboard enables course-level drill-down and mentor assignment. The Mentor and Advisor dashboard presents individual student risk cards with SHAP-driven explanations and intervention logging capability. All three dashboards are connected to a push notification and email alert engine that triggers outreach when risk thresholds are breached.

Stage 9 — Institutional Actors: The pipeline terminates with the three human actors who consume the system's outputs: the University Administrator, the School Administrator, and the Academic Mentor or Advisor. This human-in-the-loop design ensures that the platform functions as a decision-support tool that augments rather than replaces professional academic judgment.

Taken together, the framework operationalizes the transition from Strathmore University's current reactive support paradigm — where students are identified only after breaching academic thresholds — to a proactive, evidence-driven intervention model in which at-risk students are flagged weeks before failure occurs, and advisors are equipped with the specific reasons behind each risk classification.

Summary and Knowledge Gaps

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